

Animal Farm Program

1. Algorithm

Algorithm to Display Animal Details

Step 1: Start the program.

Step 2: Define a base class Animal with three methods:

- stay()
- food()
- sound()

Step 3: Create three subclasses of Animal:

- Cow
- Pig
- Horse

Step 4: Override the stay(), food(), and sound() methods in each subclass according to the characteristics of that animal.

Step 5: In the main() method, create objects of:

- Cow
- Pig
- Horse

Step 6: Display a heading indicating that animal details are being displayed.

Step 7: Call the stay(), food(), and sound() methods using the Cow object.

Step 8: Call the stay(), food(), and sound() methods using the Pig object.

Step 9: Call the stay(), food(), and sound() methods using the Horse object.

Step 10: Display the corresponding details of each animal.

Step 11: Stop the program.

2. Execution Steps

Step 1: Create the Java Class

Create the class:

AnimalFarm

inside the observation package.

Step 2: Create the Parent Class

Create the Animal class containing:

stay()

food()

sound()

These methods contain the general behavior of animals.

Step 3: Create Child Classes

Create three subclasses:

Cow extends Animal

Pig extends Animal

Horse extends Animal

Each subclass overrides the three methods to provide its own behavior.

Step 4: Create Objects

Inside main():

Cow c = new Cow();

Pig p = new Pig();

Horse h = new Horse();

Step 5: Execute the Cow Methods

The program calls:

c.stay()

c.food()

c.sound()

Output:

cows live in the farm

cows eat grass

cows do a sounds like Moooo...

Step 6: Execute the Pig Methods

The program calls:

p.stay()

p.food()

p.sound()

Output:

pigs live in the bushes

pigs eat fruits,vegetables

pigs do a sounds like oink...

Step 7: Execute the Horse Methods

The program calls:

h.stay()

h.food()

h.sound()

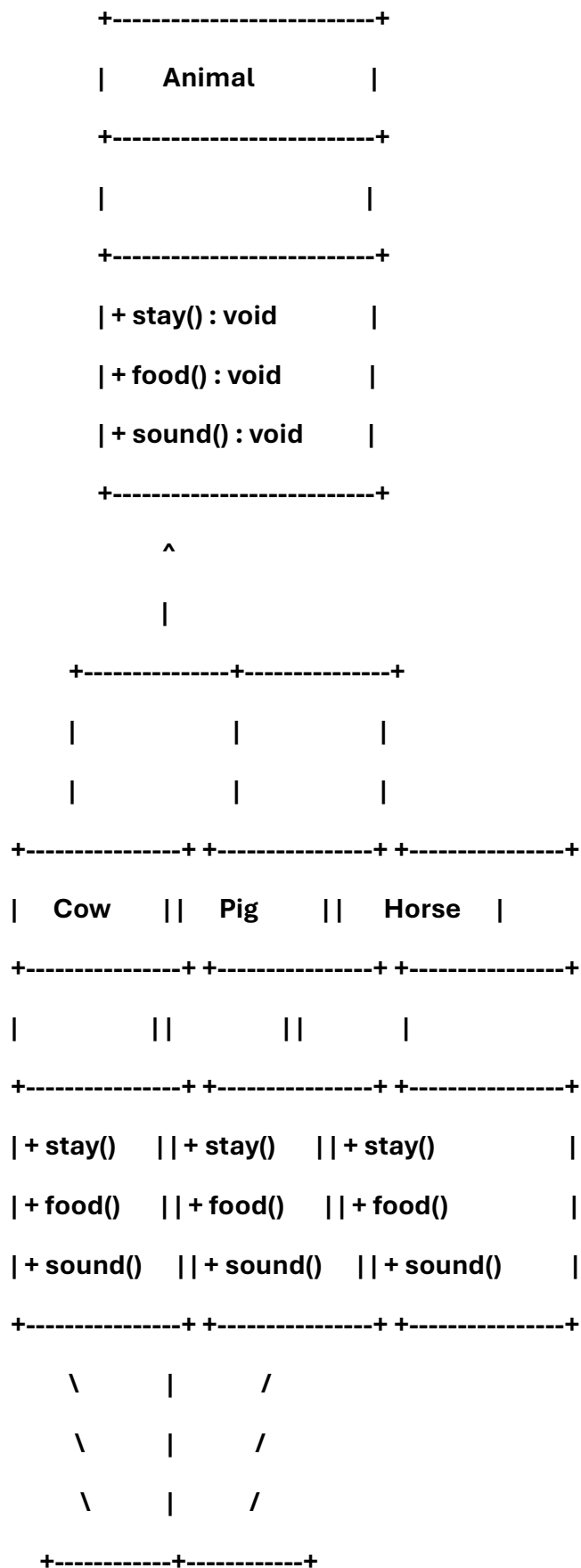
Output:

horse live in the horsefarm

horse eat grass,hay

horse do a sounds called whinny...

3. UML Class Diagram



```

      |
      v
+-----+
|  AnimalFarm  |
+-----+
|               |
+-----+
| + main(String[]) : void |
+-----+

```

Source Code:

```

package observation;

import java.util.Scanner;

class Animal{
    void stay() {
        System.out.println("animals do live outside");
    }
    void food() {
        System.out.println("animmals eat food");
    }
    void sound() {
        System.out.println("animals do sounds");
    }
}

class Cow extends Animal{
    void stay(){System.out.println("cows live in the farm");
    }
    void food() {
        System.out.println("cows eat grass");
    }
}

```

```

    }

    void sound() {
        System.out.println("cows do a sounds like Mooo...");
    }
}

class Pig extends Animal{
    void stay(){System.out.println("pigs live in the bushes");
    }
    void food() {
        System.out.println("pigs eat fruits,vegetables");
    }
    void sound() {
        System.out.println("pigs do a sounds like oink...");
    }
}

class Horse extends Animal{
    void stay(){System.out.println("horse live in the horsefarm");
    }
    void food() {
        System.out.println("horse eat grass,hay");
    }
    void sound() {
        System.out.println("horse do a sounds called whinny...");
    }
}

public class AnimalFarm {
    public static void main(String[] args) {
        Cow c=new Cow();
    }
}

```

```
Pig p=new Pig();

Horse h=new Horse();

System.out.println("the cow,pig,horse details are shown below");

c.stay();

c.food();

c.sound();

p.stay();

p.food();

p.sound();

h.stay();

h.food();

h.sound();

}

}
```

Output:

```
<terminated> AnimalFarm [Java Application] C:\Users\chinnu\OneDrive\Documents\eclipse-jee-2026-03-R-win32-
the cow,pig,horse details are shown below
cows live in the farm
cows eat grass
cows do a sounds like Moo...
pigs live in the bushes
pigs eat fruits,vegetables
pigs do a sounds like oink...
horse live in the horsefarm
horse eat grass,hay
horse do a sounds called whinny...
```